

RANI ANDRIANI TUNGGAL

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Biochemistry graduate from IPB University with hands-on experience in sample preparation, chemical and biochemical analysis, and routine laboratory testing. Experienced in UV-Vis spectrophotometry, gravimetric analysis, pH measurement, biochemical assays, FTIR, TLC, and laboratory reagent preparation, with experience supporting the analysis of 100+ samples. Supported by academic training in microbiology, industrial microbiology, analytical biochemistry, industrial biochemistry, and laboratory safety. Detail-oriented and analytical, with a strong interest in Quality Control and pharmaceutical laboratory operations.

EDUCATION

IPB University - Indonesia

August 2022 – July 2026

Bachelor of Degree of Science in Biochemistry, 3.78/4.00

Undergraduate Thesis:

“Structural Dynamics Analysis of Arginine Tripeptide as a Potential Matrix Metalloproteinase-1 Inhibitor Candidate”

Relevant Coursework: Analytical Biochemistry, Microbiology, Industrial Microbiology, Biochemistry Industry, Laboratory Techniques and Safety

Achievements:

- 1st Place Winner, Scientific Essay Competition NEXT-G (2025) | Team Leader
- 2nd Place Winner, Scientific Writing Competition 13th AIC (2025) | Presenter & 3D Prototype Designer
- First Author, Scientific Publication in Agrokreatif: Journal of Community Service (Sinta 3) (2025)
- 3rd Place Winner, Business of The Year, PMW IPB (2024) | Product Development
- National Finalist, 37th National Student Scientific Week (PIMNAS, PKM-PM) (2024) | Team Leader
- Funding Recipient, PKM RSH (2023) | Secretary

WORK EXPERIENCES

Department of Biochemistry - IPB University

August 2025 – December 2025

Biomolecular Characterization Laboratory Assistant

- Supervised 12 undergraduate students during biomolecular characterization laboratory sessions
- Supported experiments on starch and lipid hydrolysis, protein and nucleic acid denaturation, and biomolecule characterization
- Assisted in the operation of laboratory instruments, including UV-Vis spectrophotometer, PCR machine, and agarose gel electrophoresis system
- Evaluated undergraduate students practical reports and performance

Department of Biochemistry - IPB University

August 2025 – December 2025

Nucleic Acid and Protein Engineering Laboratory Assistant

- Supervised 11 undergraduate students during gene isolation and cloning experiments involving the amylase-coding gene from *Bacillus licheniformis*
- Facilitated molecular biology procedures including genomic DNA extraction, PCR-based cloning, agarose gel electrophoresis, and plasmid DNA digestion
- Supported laboratory experiments through the operation of PCR, NanoDrop spectrophotometer, and agarose gel electrophoresis systems
- Evaluated undergraduate students practical reports and performance

Universitas Terbuka Bogor, Department of Biochemistry - IPB University

November 2025

Laboratory Assistant-Daily Worker

- Supervised 7 undergraduate students during biochemistry laboratory practical sessions involving titration, UV-Vis spectrophotometry, biochemical assays, paper chromatography, and DNA analysis
- Demonstrated proper laboratory procedures and instrument operation, including agarose gel electrophoresis techniques

Indonesian Oil Palm Research Institute - Bogor

July 2025 – August 2025

Biomolecule Laboratory Assistant-Internship

- Assisted in the analysis of 30+ salak leaf DNA samples using gradient PCR and agarose gel electrophoresis
- Assisted in the analysis of 41 oil palm leaf DNA samples using PCR and polyacrylamide gel electrophoresis (PAGE)
- Supported oil palm tissue culture research through the preparation, sterilization, and observation of 100 oil palm sprouts
- Operated a viscometer to measure the viscosity of submitted soap samples for analytical testing

Indonesian Oil Palm Research Institute - Bogor

December 2024 – January 2025

Biomolecule Laboratory Assistant-Internship

- Performed moisture and phosphate analyses on 30+ plant samples using gravimetric methods and UV-Vis spectrophotometry
- Conducted protein and xylanase activity assays on 30+ submitted samples while ensuring accurate and reliable data generation
- Prepared laboratory equipment, buffers, and chemical reagents and operated a pH meter to support routine analytical testing

Tropical Biopharmaca Research Center - Bogor

June 2024 – August 2024

Laboratory Assistant-Internship

- Prepared and extracted natural product samples using sonication and rotary evaporation, followed by proximate and phytochemical analyses of submitted samples using UV-Vis spectrophotometry
- Conducted chemical identification and characterization of 10+ samples using FTIR spectroscopy and thin-layer chromatography (TLC)
- Performed laboratory procedures under supervisor guidance

PROJECTS

Academic Project - Agricultural Biochemistry

August 2024 – December 2024

The Effect of Potassium Fertilizer on Growth, Primary and Secondary Metabolite Contents, and Antioxidant Capacity of *Coleus amboinicus*

- Conducted a controlled experimental study to evaluate the effects of potassium treatment on plant growth, chlorophyll, starch, and primary metabolite content
- Quantified secondary metabolites and antioxidant capacity using UV-Vis spectrophotometric assays, including total phenolic, flavonoid, CUPRAC, and ABTS assays
- Analyzed experimental results to evaluate treatment effects on biochemical and antioxidant parameters

Academic Project - Tropical Biopharmaca Research Center

June 2024 – August 2024

Thin Layer Chromatography Fingerprint Analysis of *Simplicia X*

- Performed TLC fingerprint analysis using CAMAG Linomat 5, TLC Silica Gel, and UV detection
- Evaluated analytical performance through visual stability, specificity, and precision testing by comparing chromatographic profiles and R_f values

ORGANIZATIONAL EXPERIENCES

BEM FMIPA - IPB University

January 2024 – December 2024

Staff of Business and Partnership Bureau

- Supported G-astip and Tutor BEM FMIPA programs, including financial and profit-sharing records using Microsoft Excel
- Designed promotional materials using Canva, achieved over 20+ G-astip customers and 5 Tutor participants

Persekutuan Mahasiswa Kristen - IPB University

January 2024 – December 2024

Secretary of Special Services Commission

- Managed member and alumni databases using Google Forms
- Created and distributed monthly activity newsletters using Canva

KIP Kuliah Mengabdi - IPB University

January 2023 – January 2024

Mentor

- Mentored 10 KIP-K scholarship students in understanding the KIP-K system and preparing community service activities at their former or nearby schools in Bogor
- Prepared mentoring materials using Microsoft PowerPoint and supported student registration via Google Forms

SKILLS

- **Hard Skills:** Laboratory & Analytical Skills (sample preparation, chemical & biochemical analysis, microbiological analysis, media preparation, laboratory sterilization, total plate count (TPC), gravimetric analysis, and buffer & reagent preparation), Laboratory Instruments (centrifuge, pH meter, viscometer, sonicator, autoclave, micropipette, UV-Vis, NanoDrop, PCR, Agarose Gel Electrophoresis, SDS-PAGE, TLC, FTIR), Microsoft Office (Word, Powerpoint, Excel), Google Workspace
- **Soft Skills:** Teamwork & Collaboration, Attention to Detail, Analytical Thinking, Problem Solving, and Leadership
- **Language:** native Indonesia, basic English
- **Certifications:** Healthcare Regulatory Training (Drugs and Cosmetics) based on BPOM Regulations – Ioda academy, Patent Specification Writing and Drafting Training – DRI IPB University